LEF MODEL · CHAPTER 5

# Chapter 5: The Pear and the Stefan Release — the Bridge That Carries the Moon and Tears Exactly at the Corotation Line

Based on Paper 03 — LEF Mechanism in Practice (Part C: §C.1–C.6)

01 · THE MYSTERY

## The mystery: how do you hold a whole moon at latitude 22.5° — and detach it without leaving a tail?

In the previous chapters we saw how the internal engine pushes foamed magma upward (Chapter 3), and how the outer sial crust wraps the rising body as a dome (Chapter 4). But as soon as we look at the engineering picture of **the connection between the Earth and the Moon** toward the end of the climb, a double and complex riddle appears:

1. **The enormous sideways force at latitude 22.5°:** the Moon does not sit on the equator, but at latitude $\lambda = 22.5^\circ$ (17,287 km from the Earth's centre, with a 5.77-hour day). While on the corotation line the radial pull balances exactly ($a_r = 0$), the centrifugal push of the rotation points away from the axis of rotation — and so it has a sideways component ($a_\lambda = 0.559\ \mathrm{m/s^2}$) that pulls the Moon strongly toward the equator! This is an enormous sideways shear load of **4.06 × 10²² newtons**. How can a hot, molten "neck" hold such a force without bending and sliding to the equator?

2. **The clean break without a narrow neck:** in familiar fluid mechanics, when a drop detaches from a syringe or a tap, it stretches into a thin thread and tears — leaving part of the material behind. Had the Moon separated in this

way, it would have been dragged along with a distorted "tail" or broken apart. How is it released as a round, whole and rigid body, at a precise distance of about **210 km** beyond the corotation line?

Part C of Paper 03 solves both riddles together: the connection is not a thin "neck" but **a wide, hot contact disc 3,000 km across**, sitting on the broad back of a gravitationally solid pear; and the separation is not a brittle fracture of a solid, but **a viscous adhesion phenomenon (Stefan adhesion)** in which a thin layer of magma gradually opens up the moment the centre of mass crosses the corotation line!

02 · AN IMAGE

## An image: two glass plates with a drop of honey between them

To understand how a wide, hot contact disc separates, imagine a simple kitchen experiment: take two flat glass plates and put a thin layer of water or honey between them.

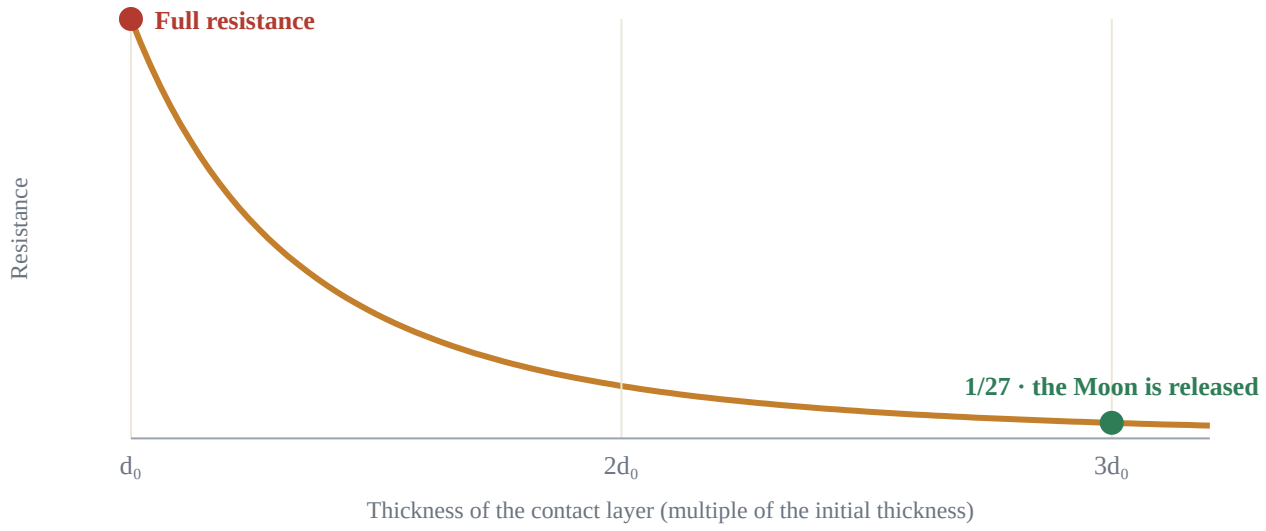
• If you try to **slide** the plates sideways over each other, you will find very little resistance.

• But if you try to **pull them straight apart**, at right angles, you will suddenly feel a strong resisting force! Why? Not because the honey or water is a solid glue, but because for the gap between them to open, new fluid must flow inward from the edges to the centre — and when the gap is very narrow, the inward flow is slow and resists the rate of opening.

And what happens if you keep pulling? As soon as the gap opens a little (say, to three times its initial thickness), the fluid flows in easily, **the resistance collapses all at once (it falls 27-fold!)**, and the two plates come completely apart!

In physics this phenomenon is called **the Stefan release**. The hot contact layer between the pear and the Moon behaves exactly like that layer of honey between two huge plates 3,000 km wide: as long as the Moon's centre of mass is inside the corotation line, gravity presses the plates together; the moment the centre of mass

crosses the corotation line outward, the force reverses into tension, the contact layer opens slowly at first and collapses all at once about **210 km** beyond the line!




The Stefan release: the resistance to opening the layer falls as the third power of its thickness; at three times the thickness it is 27 times smaller — a displacement of about 210 km.

03 · SCIENCE AND MODEL

## Science and model: keeping the concepts apart — force, stress and anchoring

To avoid common errors in calculating planetary loads, Paper 03 carefully separates three basic concepts: **force** (the total push or pull, in newtons), **stress** (the force divided by the area that carries it, in gigapascals), and **deformation**. Here are the three main conclusions that follow:

1. **The upper dome does not carry the source pressure of 150 GPa:** in hydrostatic equilibrium, most of the deep pressure is used to lift the column of rock below it. The self-pressure at the Moon's centre ($R_b = 1,890$ km, density 2,600 kg/m³) is only $p_c \simeq 3.37$ GPa — forty times less than the source pressure!

2. **Why a narrow neck fails — and a wide contact disc (3,000 km) wins:** the sideways load of 4.06 × 10²² newtons at latitude 22.5° is spread over the area of the connection. Had the connection been a neck with a radius of 1,000 km, the shear stress on it would have been 12.92 GPa (and for a 500-

km neck — 46.04 GPa), a stress that would make hot material flow and smear toward the equator. By contrast, on the wide contact disc of stage 3 (3,000 km across, radius $R_c = 1,500$ km), the average stress falls to **5.74 GPa**, and it is carried by the broad structure of the pear beneath, which is held by the Earth's own gravity!

3. **Why the Moon did not slide to the equator on the way up:** two anchoring systems kept the rising body from sliding toward the equator under the sideways centrifugal force: in the wide-dome stage (6,000 km), the body was anchored inside the opening in the stiff sial shell ($\eta \gtrsim 10^{23}$ Pa·s, the same shell that holds 100-km craters without collapse); and in the outward-rounding stage (3,000 km), the sideways load was transferred directly to the massive pear beneath.

A double separation at the same time (peeling + pulling)

The parting of the Moon from the Earth took place in two complementary ways: along the outer ring of the lower dome (from 6,000 to 3,000 km), the sial layer on which the body sat **gradually peeled off** along the "lubricated" boundary above the sima and joined the Moon (creating the 31°–70° belt); while in the central contact disc (3,000 km), the extreme heat melted and removed the sial, and the hot layer was pulled and opened in the **Stefan release** right after the corotation line was crossed!

04 · THE MECHANISM IN WORDS

## The mechanism in words: the Stefan release and the 210-km margin that explains two lunar facts at once

What exactly happens at the moment the Moon's centre of mass crosses the corotation line (standing at latitude 22.5°, at a distance of 17,287 km, with a 5.77-hour day)?

1. **The turning point at the centre of mass:** the Moon's far side crosses the corotation line earlier during the climb, but as long as the centre of mass is

inside the line, the inner part of the Moon is (gravitationally) heavier than the outer part, and the contact stays under compression. The moment the centre of mass crosses the line, the sum of the forces reverses into outward tension ($F_{\text{net}} = m \cdot 3\omega^2 \Delta r$).

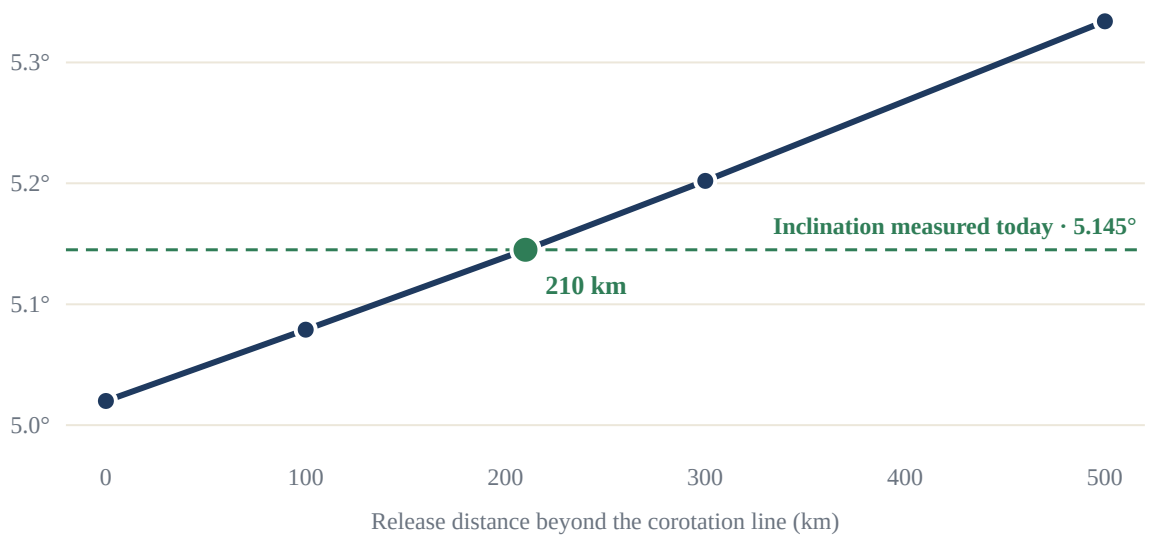
2. **The collapse of the Stefan layer ($d = 3d_0$):** the hot contact layer (initial thickness $d_0 \approx 105$ km, viscosity $\eta_c$) at first resists the rate of opening, but as the gap opens the resistance drops as the third power of the gap ($d^{-3}$). When the gap reaches three times its initial thickness ($d = 3d_0$, that is, a displacement of $\Delta r_{\text{release}} \simeq 2d_0 \approx 210$ km beyond the corotation line, when about 58% of the Moon's volume has already crossed the line), the resistance falls by a factor of 27 and the Moon is set free.

3. **Rotational coupling without premature tearing:** throughout the whole climb (at a rate of about 1 km per year), the same contact disc gives the Moon the torque needed to keep it in corotation — at a tiny shear stress of only about 0.2 kPa (about one five-hundredth of the air pressure at sea level!).

**The double proof from today's lunar orbit (§C.5a):**

Why was the Moon released about **210 km** beyond the corotation line, and not exactly on the line itself? It turns out that release exactly on the line (0 km) would put the Moon into a perfectly circular orbit ($e_0 = 0$), and the Earth's tide keeps a circular orbit circular forever — meaning that the eccentricity (the elliptical flattening) of the Moon's orbit today (e = 0.0549) would never have arisen! The computational search in Paper 03 finds that two independent tests meet at exactly the same release distance:

- **The orbital-inclination test:** release at latitude 22.5° about 210 km beyond the line ($a_0 = 18,163$ km, $e_0 = 0.037$) gives, over the orbital evolution, a final inclination of **5.145°** — the measured angle of the Moon's orbit today!
- **The eccentricity test:** exactly the same margin (≈ 210 km) reproduces, for a young, hot Moon (A ≈ 3.0), the eccentricity measured today — **0.0549**!




The final orbital inclination by release distance (Paper 03, Part C): only release about 210 km beyond the line gives 5.145° — and at the same distance also the eccentricity of 0.0549.

05 · THE KEY EQUATIONS

## Academic explanation and the key equations

The loads acting on the connection at latitude $\lambda$ (the radial $a_r$ and the sideways $a_\lambda$) and the resisting force of the Stefan layer of thickness d and radius $R_c = 1,500$ km are expressed in the following equations:

$$a_r = \frac{G(M_E + m)}{r^2} - \omega^2 r \cos^2\lambda$$

$$a_\lambda = \omega^2 r \sin\lambda \cos\lambda$$

$$F_{\text{conn}} = m_r \sqrt{a_r^2 + a_\lambda^2}$$

$$F_{\text{Stefan}} = \frac{3\pi\eta_c R_c^4 \dot{d}}{2d^3} = m(3\omega^2)(d - d_0) \quad \Longrightarrow \quad \Delta r_{\text{release}} \simeq 2d_0 \approx 210 \text{ km}$$

At the adopted release point ($\lambda = 22.5^\circ$, a 5.77-hour day), the radius of the corotation line is $r = r_{\text{co}} \cos^{-2/3}\lambda = 17,287$ km, and the Moon's centre at the

moment of full detachment stands at **17,496 km** (and for the equator: the line at 16,399 km and release at 16,610 km). The release leaves an almost circular orbit with an initial eccentricity $e_0 \simeq 0.037$ and a semi-major axis $a_0 = 18,163$ km, which are passed as direct input to Paper 04 (*The Orbit*).

06 · THE SIGNATURE

# The signature in reality: the last connection at Mare Imbrium and the spatula ridges

The Stefan release and the two-stage parting have tangible fingerprints on the face of the Moon itself:

- **The order of separation of the great maria:** along the peeling ring, **Mare Serenitatis** was exposed and peeled off first, while **Mare Imbrium** (lying deep inside the 3,000-km contact disc) served as the last internal feeding conduit and stayed attached to the pear until the moment of final detachment! Once Imbrium was cut off, the Earth's internal pressure on the Moon stopped, and the great maria of the near side slowly froze as smooth surfaces like a frozen lake.
- **The spatula ridges:** when the Moon's centre of mass crossed the corotation line and the outward pulling forces overcame the resistance of the hot, soft contact, the strong outward pull stretched and scored the surface of the detachment zone, creating the characteristic **spatula ridges**.

## The lab box: the forces of the connection, the Stefan release and the search for the release distance (Paper 03, Part C)

| Distance beyond the corotation line ($\Delta r$) | Volume of the Moon past the line | Initial semi-major axis ($a_0$) | Initial eccentricity ($e_0$) | Final orbital inclination (today) | Meaning and result of the test |
|---|---|---|---|---|---|
| **0 km (exactly on the line)** | 50% | 17,287 km | 0.000 | 5.020° | A circular orbit forever — does not reproduce e = 0.0549 |

| | | | | | |
|---|---|---|---|---|---|
| **100 km** | 54% | 17,696 km | 0.017 | 5.079° | Inclination too low; eccentricity too low (0.013 at A=3) |
| **≈ 210 km (the adopted state)** | **58%** | **18,163 km** | **0.037** | **5.145° (measured!)** | **Reproduces at once the inclination of 5.145° and e ≈ 0.0549** |
| **300 km** | 62% | 18,571 km | 0.053 | 5.202° | Inclination and eccentricity (0.11) too high |
| **500 km** | 69% | 19,531 km | 0.089 | 5.334° | A significant departure from the Moon's data today |

| Release latitude | Length of the Earth's day | The corotation line | The Moon's centre at full release | Load on the connection at the moment of release |
|---|---|---|---|---|
| **0° (the equator)** | 5.77 hours | 16,399 km | ~ 16,610 km | Full radial balance; only negligible Stefan resistance |
| **22.5° (the adopted route)** | **5.77 hours** | **17,287 km** | **17,496 km** | $a_r = 0$**; sideways load** $F_{\text{conn}} = 4.06 \times 10^{22}$ N **(5.74 GPa)** |

07 · NEXT

# The door ahead

We have seen how the Moon is released as a rigid, whole sphere exactly 210 km beyond the corotation line, with no narrow neck and without leaving a tail of fragments in space. But what happened to the enormous body left behind beneath the contact disc — **the pear itself**, about 8,700 km long and nine to twelve Moon masses in mass?

In the next chapter — **Chapter 6: The Fate of the Pear and the Scars of Reality — from the Pacific Basin to the Earth's Core** (Paper 03, Parts C.7–C.11, D and E) — we will discover why the whole pear was drawn back into the Earth, where its

footprint lies on the floor of the Pacific Ocean and at the bottom of the mantle, and how it even stretched the early Earth's core!

Chapter 6

The Fate of the Pear and the Scars of Reality — from the Pacific Basin to the Earth's Core →

← Chapter 4 · Prologue

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